

Traffic Assignment

Route choice modeling in transportation networks

Lewis Lehe

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Four-Stage Planning Models

- Transportation planning traditionally uses a four-stage approach:
 1. Trip generation: How many trips are produced/attracted?
 2. Trip distribution: Where do trips go?
 3. Modal split: Which transportation mode is used?
 4. **Traffic assignment**: Which routes are taken?
- Traffic assignment determines how travelers choose routes through a network
- Key challenge: Travel time on a link depends on its flow (congestion)

Bureau of Public Roads (BPR) Function

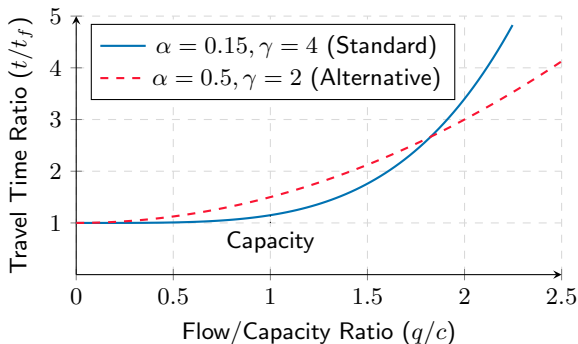
- The BPR function relates travel time to flow on a link:

$$t(q) = t_f \left[1 + \alpha \left(\frac{q}{c} \right)^\gamma \right] \quad (1)$$

- Where:

- $t(q)$ = travel time at flow q
- t_f = free-flow travel time
- q = flow on the link
- c = capacity of the link
- α, γ = parameters (typically $\alpha = 0.15$ and $\gamma = 4$)

BPR Function Example



- At capacity ($q/c = 1$), travel time is $(1 + \alpha)$ times the free-flow travel time

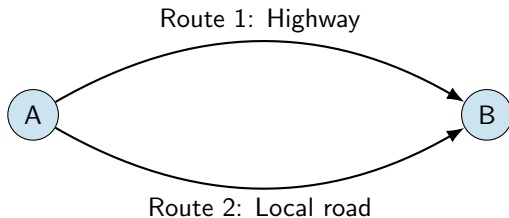
Wardrop's First Principle: User Equilibrium (UE)

- John Glen Wardrop (1952) proposed two principles for traffic assignment
- **First principle (User Equilibrium)**: "The journey times on all routes actually used are equal, and less than those which would be experienced by a single vehicle on any unused route."
- In other words:
 - No traveler can unilaterally reduce their travel time by changing routes
 - All used paths between an origin and destination have equal travel times
 - If there are unused paths, they have higher travel times than used paths
- This is a Nash equilibrium: each traveler makes an optimal decision given what others are doing

Simple Network Example

Two-Route Network Example

- Consider a simple network with one origin (A) and one destination (B)
- Two routes connect A to B:
 - Route 1: Highway with high capacity but longer distance
 - Route 2: Local road with lower capacity but shorter distance



- Travel time functions:

$$t_1(q_1) = 2 + \frac{q_1}{6} \quad (\text{tu}) \qquad t_2(q_2) = 1 + \frac{q_2}{2} \quad (\text{tu}) \qquad (2)$$

- Total demand: $Q = q_1 + q_2 = 3$ vehicles

Finding the User Equilibrium

- In equilibrium, the travel time on each route is the same (if people take both routes).
- Make the travel time on each route equal:

$$t_1(q_1) = t_2(q_2) \quad (3)$$

$$2 + \frac{q_1}{6} = 1 + \frac{q_2}{2} \quad (4)$$

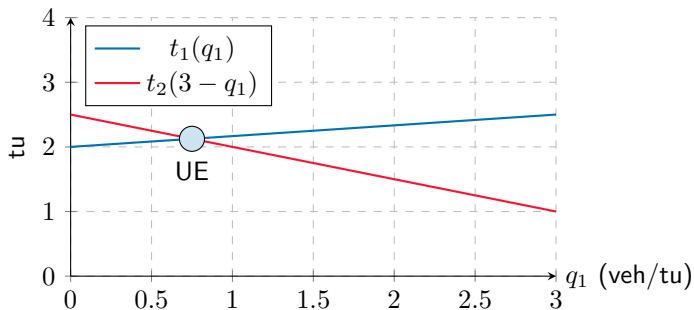
- With the constraint $Q = q_1 + q_2 = 3$, we can substitute $q_2 = 3 - q_1$:

$$2 + \frac{q_1}{6} = 1 + \frac{3 - q_1}{2} \quad (5)$$

- Solving this equation:

$$q_1 = 0.75 \qquad q_2 = 2.25 \qquad t_1 = t_2 = 2.125 \quad (6)$$

User Equilibrium: Graphical Approach



- The equilibrium occurs where $t_1(q_1) = t_2(q_2)$, i.e., where the two curves intersect
- Both routes have a travel time of 2.125 tu

Corner solutions

- Suppose $Q = q_1 + q_2 = 3$ (veh/tu) and

$$t_1(q_1) = \textcolor{red}{3} + \frac{q_1}{6} \qquad t_2(q_2) = 1 + \frac{q_2}{2} \qquad (7)$$

- Then in equilibrium

$$t_1(q_1) = t_2(Q - q_1) \qquad (8)$$

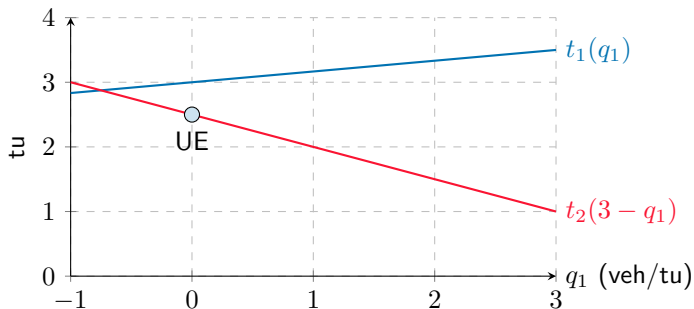
$$3 + \frac{q_1}{6} = 1 + \frac{3 - q_1}{2} \qquad (9)$$

$$- \frac{2q_1}{3} = 0.5 \qquad (10)$$

$$q_1 = -0.75 \qquad (11)$$

- **Everyone takes one route**...in this case $q_1 = 0$ and $q_2 = 3$ and $t_2(3) = 1 + 3/2 = 2.5$.

User Equilibrium: Corner Solution



- What if the intersection has $q_1 < 0$ (or $q_1 > Q$)?

System Optimum

Wardrop's Second Principle: System Optimum (SO)

- **Second principle (System Optimum):** "The average journey time is a minimum."
- This minimizes the **aggregate travel time**: $\sum_i q_i \cdot t_i(q_i)$ (veh-tu)
- For two links: $q_1 \cdot t_1(q_1) + q_2 \cdot t_2(q_2)$
- Unlike user equilibrium, **the system optimum may have different travel times on different routes**
- Individual travelers will not choose this distribution on their own

System Optimum Conditions

- Let $\phi = q_1 \cdot t_1(q_1) + q_2 \cdot t_2(q_2)$ be the **aggregate travel time**
- Since $Q = q_1 + q_2$,

$$\phi = q_1 \cdot t_1(q_1) + (Q - q_1) \cdot t_2(Q - q_1) \quad (12)$$

- To find the **system optimum** (the value of q_1 that minimizes ϕ), we take the derivative of ϕ with respect to q_1 and set it equal to zero (**First-Order Condition**):

$$\frac{d\phi}{dq_1} = t_1(q_1) + q_1 \cdot t_1'(q_1) - t_2(Q - q_1) - (Q - q_1) \cdot t_2'(Q - q_1) = 0 \quad (13)$$

Now solve for q_1 .

Finding the System Optimum

- Our example has

$$t_1(q_1) = 2 + \frac{q_1}{6} \qquad t_2(q_2) = 1 + \frac{q_2}{2} \qquad Q = 3 \qquad (14)$$

- Hence,

$$\phi = q_1 \cdot t_1(q_1) + (3 - q_1) \cdot t_2(3 - q_1) \qquad (15)$$

$$\phi = q_1 \cdot \left(2 + \frac{q_1}{6}\right) + (3 - q_1) \cdot \left(1 + \frac{3 - q_1}{2}\right) \qquad (16)$$

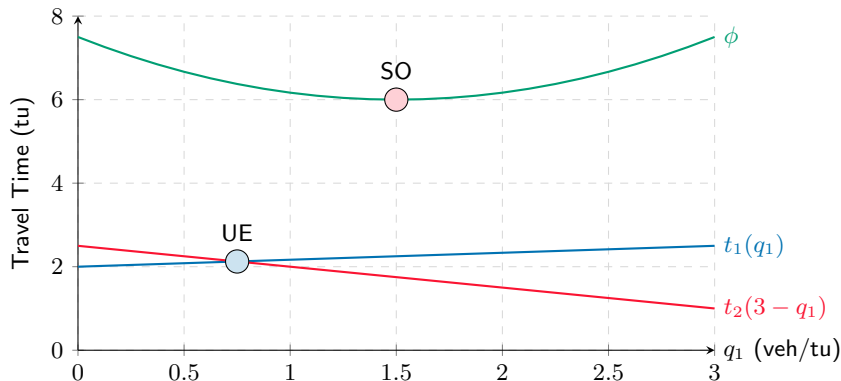
$$\phi = 2q_1 + \frac{q_1^2}{6} + (3 - q_1) + \frac{(3 - q_1)^2}{2} \qquad (17)$$

$$\frac{d\phi}{dq_1} = 2 + \frac{q_1}{3} - 1 - (3 - q_1) \qquad (18)$$

$$\frac{d\phi}{dq_1} = -2 + \frac{4}{3}q_1 = 0 \qquad (19)$$

$$q_1 = \frac{3}{2} \qquad (20)$$

System Optimum vs. User Equilibrium

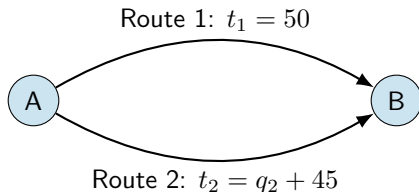


- At UE: $q_1 = 0.75$, $q_2 = 2.25$, $t_1 = t_2 = 2.125$ and $\phi = 6.375$ (veh-tu)
- At SO: $q_1 = 1.5$, $q_2 = 1.5$, $t_1 = 2.25$, $t_2 = 1.75$ and $\phi = 6$ (veh-tu)
- Note it is a coincidence that the traffic is equal in the SO here. That will not usually be true.

Traffic Assignment “Paradoxes”

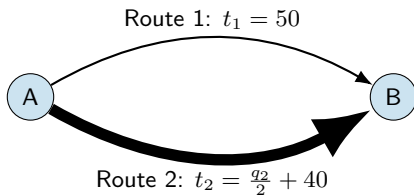
Pigou-Knight-Downs Paradox

- A classic paradox in traffic assignment (Pigou 1920, Knight 1924, Downs 1962)
- Principle: Adding capacity to a road network can leave travel times unchanged
- Also known as the “road improvement paradox” or “Downs’ paradox”



- Total demand: $Q = 30$ vehicles
- At equilibrium: $t_1 = t_2 \Rightarrow 50 = q_2 + 45 \Rightarrow q_2 = 5, q_1 = 25$
- Travel time: 50 time units; Total system time: $25 \cdot 50 + 5 \cdot 50 = 1500$

Pigou-Knight-Downs Example: After Improvement



- Route 2 is improved (widened)
- New travel time function: $t_2 = \frac{q_2}{2} + 40$
- At new equilibrium:

$$t_1 = t_2 \quad (21)$$

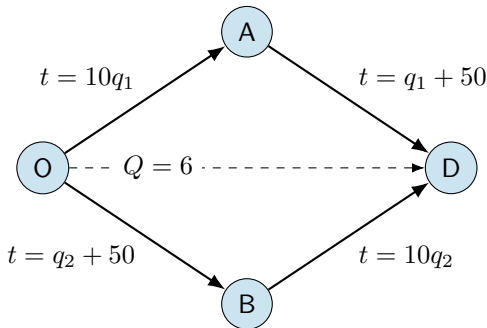
$$50 = \frac{q_2}{2} + 40 \quad (22)$$

$$q_2 = 20, q_1 = 10 \quad (23)$$

- Travel time: **still 50 time units!**
- The paradox: Despite improving route 2, equilibrium travel time stays the same!
- Benefits completely consumed by route shifting toward the improved route

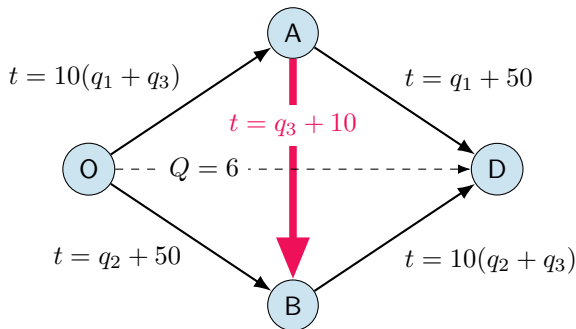
Braess Paradox: Introduction

- Discovered by Dietrich Braess in 1968
- The paradox: Adding a new link to a network can increase travel times for all users
- Contrary to intuition: more options should never make things worse
- Has been observed in real networks (Seoul, Stuttgart, New York)



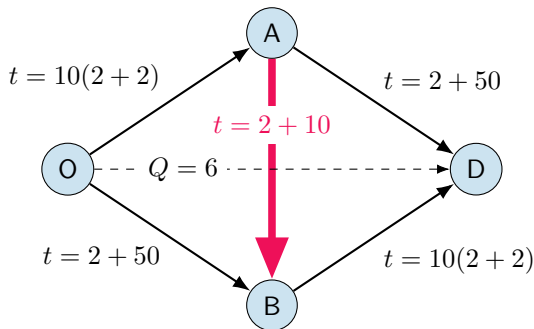
- Two routes. 1: $O \rightarrow A \rightarrow D$ and 2: $O \rightarrow B \rightarrow D$ with total demand of $Q = 6$
- At equilibrium: $q_1 = q_2 = 3$ with travel time = 83 tu

Braess Paradox: Network with New Link



- New link $A \rightarrow B$ added with $t = q_3 + 10$
- Route 3: $O \rightarrow A \rightarrow B \rightarrow D$

Braess Paradox: Network with New Link



- The new equilibrium has $q_1 = q_2 = q_3 = 2$
- The travel time is now $40 + 40 + 12 = 92$
- The improvement has made things worse.